RYERSON UNIVERSITY
DEPARTMENT OF PHYSICS

PCS 125 FALL 2006 FINAL-TERM EXAMINATION

DURATION: 150 minutes **DATE: December 5, 2006**

Student Number	**Section**	**PRINT first name**	**PRINT family name**

Circle your instructor's name.

Antimirova **Ford** **Kumaradas** **Toronov** **Szmidt**

INSTRUCTIONS

- Check that the "bubble sheet" at your desk has your name and student number on it.
- This is a closed-book exam; pens, pencils, erasers, calculators and your Ryerson ID are the only allowed items on your desk. Your Ryerson photo ID must be on your desk at all times.
- The only calculators allowed are the Sharp EL-546 and Casio FX-991**.**
- **PDAs**, **phones, and pagers must be turned off** and **out of reach.** Coats, jackets, caps, food, drinks and bags must be placed out of reach.
- The examination has a total of 70 marks and consists of 2 sections:
- **SECTION A: MULTIPLE-CHOICE QUESTIONS (#1—#17)**
  - All multiple-choice questions are worth 3 marks each.
  - We may check your work, and if your answer is not justified by it, credit will not be given. You may use the back of the pages for rough work.
  - Select the option that is numerically closest to your answer, circle it in this booklet and then copy it to the bubble sheet carefully using a pencil.
  - **Only the bubble sheet will be marked.**
  - If more than one choice is presented for a question (on either the test sheet or the bubble sheet), you may not get any credit for that question.
- **SECTION B: LONG-ANSWER QUESTIONS (#18—#20)**
  - The marks assigned to each question are shown in brackets with each question.
  - For questions that require numerical answers, make sure you include all significant steps (such as equations and their numerical substitutions) in your solution, presented in the correct order. Unjustified answers will be given little or no credit. Cross out or erase all rough work not relevant to your solution. Put a box around your final answer.
  - Write your solutions on the front of the page, below the questions. If you need more space, use the back of the page, but clearly indicate this fact on the original page, making sure that your solution cannot be confused with any rough work which may be there.

Please sign here indicating you have read and understood the above instructions.

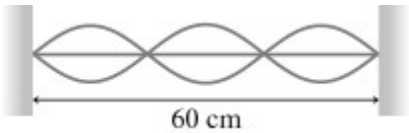
Signature: _____

DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO

Then, before you start writing, verify that you **do** have all 12 pages (otherwise inform one of the invigilators).

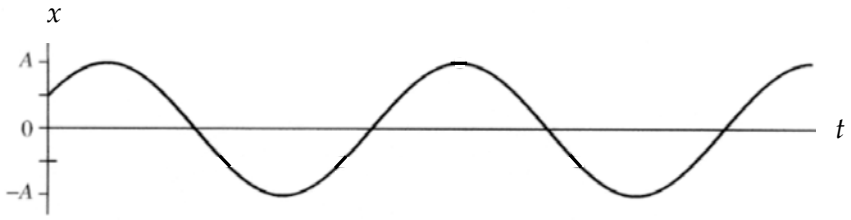
SECTION A: MULTIPLE-CHOICE QUESTIONS

1. A standing wave is oscillating at 950 Hz on a string, as shown in the figure. What is the wave speed?



a) 380 m/s b) 190 m/s c) 290 m/s d) 570 m/s e) 460 m/s

2. The graph below represents a particle in simple harmonic motion. What is the phase constant, ϕ_0 ?



a) 0 b) –2.09 rad c) 1.05 rad d) 2.09 rad e) –1.05 rad

3. A diffraction grating with 4000 lines/cm is illuminated by light from the sun. The solar spectrum is spread out on a white wall across the room. At what angle from the centre line does blue light (400 nm) appear?

a) 0° b) 12.7° c) 8.3° d) 15.1° e) 9.2°

SECTION A: MULTIPLE-CHOICE QUESTIONS

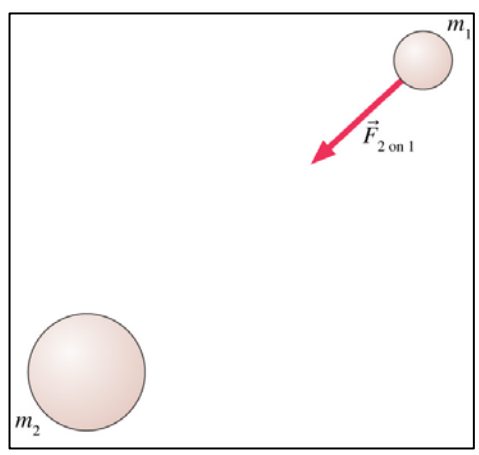
4. Monochromatic light (λ = 500 nm) is incident on a soap bubble (n = 1.40). How thin is the bubble (in nm) if destructive interference occurs in the reflected light?

   a) 102 b) 179 c) 54 d) 1 e) 89

5. In Young's double-slit experiment, two slits separated by 0.050 mm are illuminated with green light (λ = 540 nm). How many bands of bright light are on the screen between the central maximum and the 12 cm position? (The distance between the double slits and the screen is 1.0 m).

   a) 15 b) 11 c) 1 d) 22 e) 0

6. The figure shows a binary star system. The mass of star 2 is twice the mass of star 1. Compared to $\vec{F}_{2\text{ on }1}$, the magnitude of the force $\vec{F}_{1\text{ on }2}$ is?



   a) one quater as big b) half as big c) twice as big
   d) four times as big e) the same

SECTION A: MULTIPLE-CHOICE QUESTIONS

7. Two stars of masses M and $6M$ are separated by a distance D. Determine the distance, measured from the star of mass M along the line containing the centres of the two stars, at which the net gravitational field would be zero.

   a) 0.41 D b) 0.33 D c) 0.37 D d) 0.29 D e) 0.14 D

8. A satellite of mass m circles a planet of mass M and radius R in an orbit at a height $2R$ above the surface of the planet. What minimum energy is required to change the orbit to one for which the height of the satellite is $3R$ above the surface of the planet?

   a) $\frac{GmM}{24R}$ b) $\frac{GmM}{15R}$ c) $\frac{GmM}{12R}$ d) $\frac{GmM}{21R}$ e) $\frac{GmM}{5R}$

9. Consider an object that is released from rest at a distance h above the surface of a planet of mass M and radius R ($R < h$). With what speed will the object strike the surface of the planet? Assume the planet has no atmosphere.

   a) $\left[\frac{2GM}{h}\right]^{1/2}$ b) $\left[\frac{GM}{h}\right]^{1/2}$ c) $\left[\frac{2GM}{R+h}\right]^{1/2}$ d) $\left[\frac{2GMh}{R(R+h)}\right]^{1/2}$ e) $\left[\frac{GMh}{R(R+h)}\right]^{1/2}$

SECTION A: MULTIPLE-CHOICE QUESTIONS

10. Which (if any) of the following 3 statements are true:
    (i) Electric field lines cross at points where $\vec{E} = 0$.
    (ii) A positively charged particle undergoes acceleration in the direction of an electric field line.
    (iii) No work is done by the electric field when a charge is moved along an equipotential.

    Select the correct option below:
    a) only (i) b) only (i) and (ii) c) only (ii) and (iii) d) only (iii) e) none

11. A charge $q_1 = 25$ nC is located at the point (–3.0 m, 0) in the *x*-*y* plane, and a charge $q_2 = -25$ nC is located at the point (+3m, 0). Find the magnitude of the electric field at the point (0, 4 m).
    a) 0 b) 10.8 N/C c) 18.0 N/C d) 54.0 N/C e) 50.4 N/C

12. Which (if any) of the following 3 statements are true:
    (i) When an electron moves from a lower electric potential to a higher electric potential, it gains potential energy.
    (ii) If 4 identical point charges are located at the corners of a square, then the electric potential at the centre of the square, due to those charges, is zero.
    (iii) Electric potential is not the same as electric potential energy.

    Select the correct option below:
    a) only (i) b) only (ii) c) only (iii) d) all e) none

SECTION A: MULTIPLE-CHOICE QUESTIONS

13. Two positive charges q_1, q_2, and a negative charge, q_3, are located at the vertices of the equilateral triangle (each side of length L). Find the work done by an external agent in moving q_3 to infinity, while q_1 and q_2 remain in their original locations.

a) $k_e \frac{q_1q_2 + q_2q_3 + q_1q_3}{L}$ b) $k_e \frac{q_1q_3 + q_2q_3}{L}$ c) $-k_e \frac{q_1q_3 + q_2q_3}{L}$

d) $k_e \frac{q_1q_2 + q_2q_3}{L}$ e) $-k_e \frac{q_1q_2 + q_2q_3}{L}$

14. At a given moment an electron is moving in the $+x$ direction in a uniform magnetic field which is in the $-z$ direction. It therefore experiences an acceleration whose direction, at that moment, is:

a) $+y$ b) $-y$ c) $+z$ d) $-z$ e) $-x$

15. A square conducting loop (of edge length = 80.0 cm) lies in the plane of this page and carries a current of 10.0 A <u>counterclockwise</u>. Determine the magnitude and direction of the magnetic field at the centre of the square.

a) 14.1 μT, into the page b) 14.1 μT, out of the page c) 20.0 μT, into the page
d) 20.0 μT, out of the page e) 15.7 μT, into the page

SECTION A: MULTIPLE-CHOICE QUESTIONS

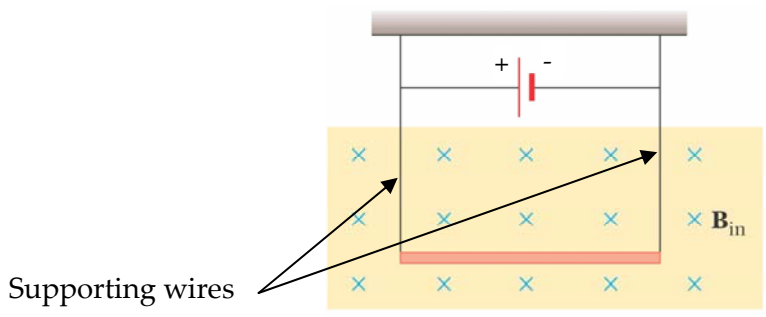
16. Which (if any) of the following 3 statements are true?
    (i) A charged particle entering a region of <u>non-uniform</u> magnetic field will move in a circle.
    (ii) The Biot-Savart law gives an expression for the magnetic force experienced by a charged particle moving in a magnetic field.
    (iii) A charged particle moving in only a magnetic field will either gain or lose kinetic energy.

    Select the correct option below:

    a) only (i) b) only (ii) c) only (iii) d) all e) none

17. A current-carrying wire in the plane of the page is suspended <u>horizontally</u> in a magnetic field of 6.0 T, directed into the page as shown in the figure. The horizontal wire conductor has mass per unit length of 0.06 kg/m. What current (in A) must exist in the wire conductor in order for the tension in the vertical supporting wires to be zero?

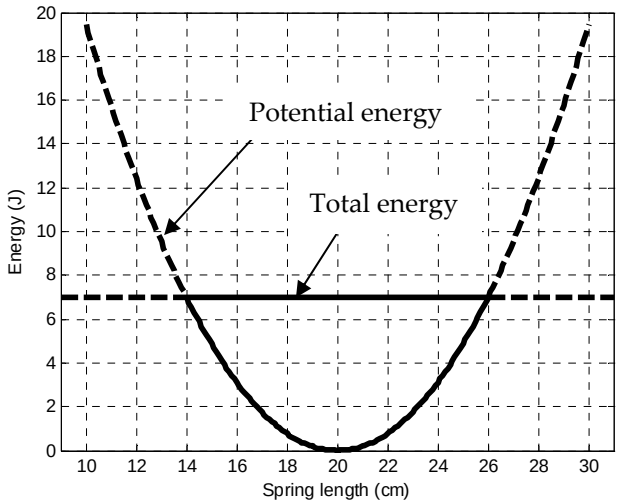
Side view



a) 10.9 b) 9.8 c) 9.8×10^{-2} d) 3.7 e) 3.7×10^{-2}

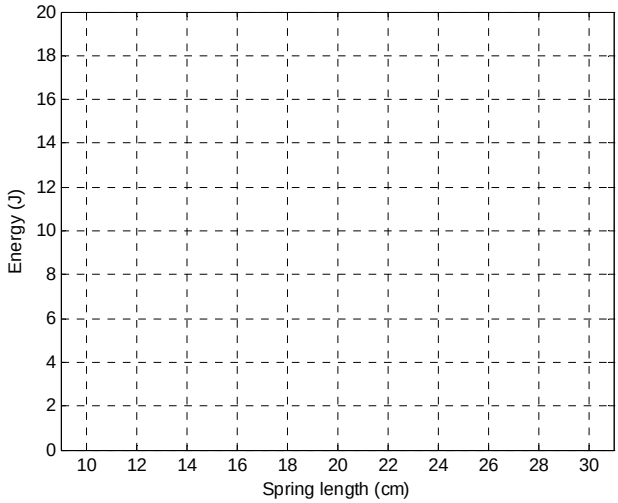
SECTION B: LONG-ANSWER QUESTIONS

18. [5 marks] The figure shows the potential energy and the total energy of a particle oscillating on a horizontal spring.



a. What is the spring's equilibrium length?

b. What is the <u>amplitude</u> of the motion? Explain how you determined it.

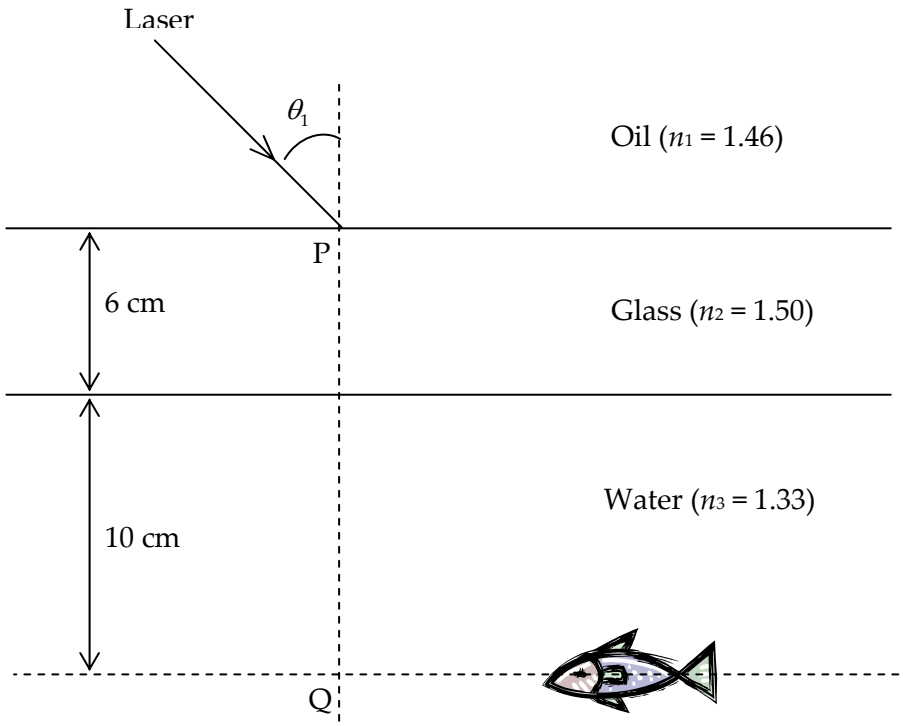


c. Draw a graph of the particle's <u>kinetic</u> energy as a function of the spring length.

d. From the first graph (at the top of this page), determine the <u>amplitude</u> of the motion (to an accuracy of ±0.5 cm) if the particle's total energy increases to 16 J?

SECTION B: LONG-ANSWER QUESTIONS

19. [5 marks] A narrow laser beam travels from oil, through a 6 cm thick glass slab, and then into water, as shown in the figure. A fish swims horizontally from the far right, towards the left, as shown.

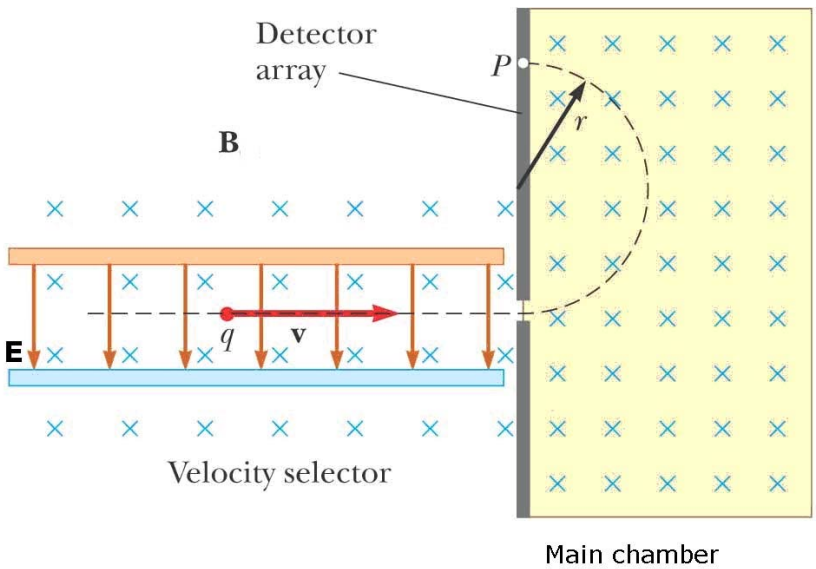


a. If the laser beam can have any angle of incidence at P, what are the minimum and maximum angles of incidence for which the fish will never see the laser beam?

b. If $\theta_1 = 30°$, at what distance from Q will the fish see the laser beam?

SECTION B: LONG-ANSWER QUESTIONS

20. [9 marks] Consider the mass spectrometer shown schematically below (in side-view). A source injects ions of only one type at various speeds horizontally into the middle of the velocity selector. The velocity selector is configured to pass only particles having a speed of 7.14×10^4 m/s into the main chamber. The magnetic field, $\vec{B}$, in the main chamber is equal to that in the velocity selector. Neglect gravitational effects when answering the following below.



a. If the magnitude of the electric field is 2500 V/m, what is the magnitude of the magnetic field, B?

b. For the path shown in the figure, are the ions positively or negatively charged?

c. What is the mass of a singly charged ion if the radius of its path in the main chamber is 27.8 cm? (Hint: the magnitude of centripetal acceleration is $\frac{v^2}{r}$.)

d. In no more than 30 words explain why a velocity selector is needed?

(More space provided on the next page.)